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Next item →

1. What strategy is proposed in this lesson to estimate a system's state vector when the process and/or sensor noises have a dc bias?

1 / 1 point

- ☐ Execute a standard Kalman filter without making any changes. It works well even when the noises have dc bias.
- ☐ Execute all the steps of a standard Kalman filter plus some additional steps to estimate the biases. Then, subtract the biases from w_k and v_k in the standard Kalman filter steps.
- ☒ Execute all the steps of a standard Kalman filter plus some additional steps to estimate the biases. Then, adjust the Kalman-filter output to compensate for the biases.
- ☐ Implement a high-pass filter to "notch out" the dc bias from the measurements and add these filter equations to the standard Kalman filter.

✔ Correct
Yes, this is correct.

2. Execute the Coursera Labs Jupyter notebook for this lesson. What is the percentage of time that the position-state error is outside of the estimation-error bounds when the filter is NOT corrected for bias? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

99.8

✔ Correct
Yes, that is correct.

3. Execute the Coursera Labs Jupyter notebook for this lesson. What is the percentage of time that the position-state error is outside of the estimation-error bounds when the filter is IS corrected for bias? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

3.2

✔ Correct
Yes. That is correct.